



Technical Description SIGMA CONTROL 2

Process Map $\geq$ piston_2.0.1

7_7601_PISTON_PA_02E

KAESER KOMPRESSOREN SE

96410 Coburg • PO Box 2143 • GERMANY • Tel. +49-(0)9561-6400 • Fax +49-(0)9561-640130

www.kaeser.com

General

1. In communication mode "send/receive" Sigma Control can monitor the master program.
This takes place with "iSlaveMsgHeader" and "iSlaveMsgFooter" that have to be returned unchanged from master.
Monitoring can be activated in menu <Communication / Ethernet/SIGMA NETWORK / Connections / SAM 4.0 / Communication error: Timeout> by entering a check mark at check box right of Timeout.
If monitoring is not required the above sequence is not used.
2. Availability or use of some signals depends on type of compressor, compressor options or use of functions at SC2.
Information see column "Availability" or "Usability" in the following tables in combination with table 3.
For further information on signal availability and use ask your local KAESER subsidiary or KAESER Headquarter Service Support Engineering, phone +49 9561 640-7979.

1. Compressor ==>>> Master Controller

Table 1: Process data

* -32768 = Value invalid or not used

4 see associated table (1.x)
 3 depends on use of the function at SC 2
 2.x see table 3
 1 always

SC 2 JSON- RPC	Description		Verfügbarkeit	Unit	min	max	Format
BDSnd							
	Alarm xAM16 - xAM1	Table 1.1	4	binary			BOOL
	Alarm xAM32 - xAM17	Table 1.1	4	binary			BOOL
	Alarm xAM48 - xAM33	Table 1.1	4	binary			BOOL
	Alarm xAM64 - xAM49	Table 1.1	4	binary			BOOL
	Alarm xAM80 - xAM65	Table 1.1	4	binary			BOOL
	Warning xWM16 - xWM1	Table 1.2	4	binary			BOOL
	Warning xWM32 - xWM17	Table 1.2	4	binary			BOOL
	Warning xWM48 - xWM33	Table 1.2	4	binary			BOOL
	Warning xWM64 - xWM49	Table 1.2	4	binary			BOOL
	Warning xWM80 - xWM65	Table 1.2	4	binary			BOOL
	Operational message xOM16 - xOM1	Table 1.3	4	binary			BOOL
	Operational message xOM32 - xOM17	Table 1.3	4	binary			BOOL
	Operational message xOM40 - xOM33	Table 1.3	4	binary			BOOL
	Binary signal xBS15 - xBS0	Table 1.4	4	binary			BOOL
	Binary signal xBS31 - xBS16	Table 1.4	4	binary			BOOL
	Binary signal xBS47 - xBS32	Table 1.4	4	binary			BOOL
	Binary signal xBS63 - xBS48	Table 1.4	4	binary			BOOL
iDV0	Software version 3rd digit / 4th digit		1				BYTE/BYTE
iDV1	Software version 1st digit / 2nd digit		1				BYTE/BYTE
iDV2	free						
iDV3	Pressure actual value pNloc		1	mbar	-400 *	20000	INT
iDV4	free						
iDV5	Block discharge temperature BDT		1	0,1 °C	-500 *	2999	INT
iDV6	free						
iDV7	free						
iDV8	free						
iDV9	free						
iDV10	Motor temperature		2.1	0,1 °C	-500 *	2999	INT
iDV11	free						
iDV12	free						
iDV13	Pressure actual value pN (pressure regulation)		1	mbar	-400 *	20000	INT
iDV14	Current drive motor starts/h		1	1			INT
iDV15	Total compressor hours run	x 1000	1	h			INT
iDV16	Total compressor hours run	x 1	1	h			INT
iDV17	Hours on load	x 1000	1	h			INT
iDV18	Hours on load	x 1	1	h			INT
iDV19	Setpoint pressure pA switching point		1	mbar			INT
iDV20	Setpoint pressure pA switching difference		1	mbar			INT
iDV21	Setpoint pressure pB switching point		1	mbar			INT
iDV22	Setpoint pressure pB switching difference		1	mbar			INT
iDV23	Nominal package pressure		1	mbar			INT
iDV24	Increased cut-out pressure pE		1	mbar			INT

1. Compressor ==>>> Master Controller

Table 1: Process data

* -32768 = Value invalid or not used

SC 2 JSON- RPC	Description	Verfügbarkeit	Unit	min	max	Format
BDSnd						
iDV25	Remaining drive motor idle time	1	s			INT
iDV26	free					
iDV27	free					
iDV28	free					
iDV29	free					
iSlaveMsg Header	Message header slave	1				WORD
iDV30	free					
iDV31	free					
iDV32	free					
iDV33	Air filter service interval	1	h			INT
iDV34	Valves service interval	1	h			INT
iDV35	Piston rings service interval	1	h			INT
iDV36	free					
iDV37	free					
iDV38	free					
iDV39	Current cut-out pressure pA (pB) switching point	1	mbar			INT
iDV40	Current cut-out pressure pA (pB) switching difference	1	mbar			INT
iDV41	Compressor model / free	1				INT
iDV42	Compressor start delay	3	s			INT
iDV43	Duty cycle	1	1%			INT
iDV44	free					
iSlaveMsg Footer	Message footer slave	1				WORD

4 see associated table (1.x)
 3 depends on use of the function at SC 2
 2.x see table 3
 1 always

1. Compressor ==>>> Master Controller

Table 1.1 : Alarms

* Summary of SC2 messages

SC 2 JSON- RPC	Messages		Availability	Description / possible cause
	BDSnd	ID Text (SC2 Display, except *)		
xAM1			3 2.x 1	depends on use of the function at SC 2 depends on model type or options, see table 3 always
xAM2	0002	Motor T ‡	2.1	Compressor drive motor overheated. (PT100)
xAM3	0003	pRV ‡	1	TÜV inspection: The activating pressure of the pressure relief valve has been exceeded.
xAM4	0004	EMERGENCY STOP	1	EMERGENCY STOP control device actuated.
xAM5				
xAM6				
xAM7				
xAM8	0008	Diagnostics group alarm	1	A diagnostic alarm occurred.
xAM9	*	Sigma Control T ‡	1	Permissible internal temperature for SIGMA CONTROL 2 MCS or IOM exceeded.
xAM10	0010	Blow-off protection ‡	1	The activating pressure of the pressure relief valve has been exceeded.
xAM11				
xAM12				
xAM13	0013	Compressor motor overload	1	Overload shut-down of the compressor drive motor.
xAM14				
xAM15	0015	BDT ‡	1	Maximum permissible block discharge temperature (BDT) exceeded.
xAM16				
xAM17				
xAM18				
xAM19				
xAM20				
xAM21	0021	Refrigeration dryer T ‡	2.T	Refrigeration dryer: Compressed air temperature too low.
xAM22				
xAM23				
xAM24				
xAM25	*	First IOM - Group alarm	1	Fault at SC 2 internal IOM
xAM26				
xAM27				
xAM28				
xAM29				
xAM30				
xAM31				
xAM32				
xAM33				

1. Compressor ==>>> Master Controller

Table 1.1 : Alarms

* Summary of SC2 messages

SC 2 JSON- RPC	Messages		Availability	Description / possible cause
	BDSnd	ID Text (SC2 Display, except *)		
xAM34	0034	Mains contactor on?	1	Mains contactor does not close.
xAM35				
xAM36				
xAM37				
xAM38				
xAM39				
xAM40	0040	Mains contactor off?	1	Mains contactor does not open.
xAM41	0041	Mains voltage ‡	2.A	Second power failure.
xAM42				
xAM43				
xAM44				
xAM45				
xAM46	*	Compressor motor EC Group alarm	2.E	EC motor: Group alarm motor electronics
xAM47	*	Compressor motor EC Communication error	2.E	Communication error between SC2 (RS485) and EC motor electronics
xAM48				
xAM49				
xAM50				
xAM51				
xAM52				
xAM53				
xAM54				
xAM55				
xAM56	0056	RD condensate drain	2.T	Refrigeration dryer: Condensate drain faulty.
xAM57				
xAM58				
xAM59				
xAM60				
xAM61				
xAM62	0062	Refrigeration dryer p ‡	2.T	Refrigeration dryer: Pressure too high in the refrigerant circuit. Safety pressure switch tripped.
xAM63				
xAM64				
xAM65				
xAM66				
xAM67	0067	Condensate drain reservoir	2.2	Tank: Condensate drain faulty
xAM68	0068	Condensate drain filter	2.3	Filter: Condensate drain faulty

3 depends on use of the function at SC 2
2.x depends on model type or options, see table 3
1 always

1. Compressor ==>>> Master Controller

Table 1.1 : Alarms

* Summary of SC2 messages

SC 2 JSON- RPC	Messages		Availability	Description / possible cause
	BDSnd	ID Text (SC2 Display, except *)		
xAM69			3	depends on use of the function at SC 2
xAM70			2.x	depends on model type or options, see table 3
xAM71			1	always
xAM72				
xAM73	0073	External message 1	3	External binary message 1
xAM74	0074	External message 2	3	External binary message 2
xAM75	0075	External message 3	3	External binary message 3
xAM76	0076	External message 4	3	External binary message 4
xAM77	0077	External message 5	3	External binary message 5
xAM78	0078	External message 6	3	External binary message 6
xAM79				
xAM80				

1. Compressor ==>>> Master Controller

Table 1.2 : Warning Messages and Maintenance Messages

* Summary of SC2 messages

SC 2 JSON- RPC BDSnd	Message		Availability	Description / possible cause
	W M ID	Warning Maintenance Text (SC2 Display, except *)		
xWM1	W 0001	Equipment number incomplete	1	The compressor equipment number consists of less than 7 characters.
xWM2	W 0002	Motor T↑	2.1	Drive motor overheating. (PT100)
xWM3	M			
xWM4	M			
xWM5	W			
xWM6	W			
xWM7	W			
xWM8	W 0008	BDT ↑	1	Maximum block discharge temperature will soon be reached.
xWM9	W			
xWM10	M			
xWM11	M			
xWM12	W			
xWM13	M			
xWM14	W			
xWM15	W			
xWM16	W			
xWM17	W			
xWM18	W			
xWM19	W			
xWM20	W			
xWM21	W 0061	Refrigeration dryer T ‡	2.T	Refrigeration dryer: Compressed air temperature too low.
xWM22	W 0062	Refrigeration dryer p ‡	2.T	Refrigeration dryer: Pressure too high in the refrigerant circuit. Safety pressure switch tripped.
xWM23	W			
xWM24	M			
xWM25	M			
xWM26	M			
xWM27	M			
xWM28	M 0028	Air filter h ‡	1	Air filter: Maintenance interval has elapsed.
xWM29	M 0029	Valves h ‡	1	Valves: Maintenance interval has elapsed.
xWM30	M 0030	Piston rings h ‡	1	Piston rings: Maintenance interval has elapsed.
xWM31	M			
xWM32	M			
xWM33	M			
xWM34	W			
xWM35	W			
xWM36	W 0036	Motor starts /h ‡	1	The permissible number of motor starts was exceeded in the last 60 minutes.
xWM37	W 0037	Motor starts /d ‡	1	The permissible number of motor starts was exceeded in the last 24 hours.
xWM38	W 0038	Blow-off protection ↑	1	The pressure relief valve's activating pressure will soon be reached.

3 depends on use of the function at SC 2
2.x depends on model type or options, see table 3
1 always

1. Compressor ==>>> Master Controller

Table 1.2 : Warning Messages and Maintenance Messages

* Summary of SC2 messages

SC 2 JSON- RPC	Message		Availability	Description / possible cause
	W M ID	Warning Maintenance Text (SC2 Display, except *)		
BDSnd			3 2.x 1	depends on use of the function at SC 2 depends on model type or options, see table 3 always
xWM39	W			
xWM40	M			
xWM41	W 0041	Mains voltage ↓	1	AIRBOX: First power failure: The machine is automatically restarted. i.Comp: Supply voltage below the limit value. The machine is off and is automatically restarted when the voltage is high enough.
xWM42	W			
xWM43	W 0043	External load signal?	3	Ambiguous external load signal: Increased cut-out pressure exceeded. The external load control has not switched to idle (off load).
xWM44	W 0044			
xWM45	W 0045	DO test active	3	The DO test is active
xWM46	W			
xWM47	W			
xWM48	M			
xWM49	M 0049	Annual maintenance	1	Appears one year after last maintenance.
xWM50	M			
xWM51	W			
xWM52	W			
xWM53	W			
xWM54	W			
xWM55	M			
xWM56	M			
xWM57	M			
xWM58	W 0058	SC2 <=> SC2 Communication error	3	SC2 to SC2 communication is interrupted.
xWM59	W 0059	Start T↓↓	1	The block temperature is too low (<-10°C / 14°F) for the machine to be operated.
xWM60	W 0060	Start T↓	1	The block temperature is too low (<+2°C / 36°F).
xWM61	*	Compressor motor EC derating	2.E	EC motor: Temperature or torque reduction (derating)
xWM62	*	Compressor motor EC power limitation	2.E	EC motor: Voltage or power limitation
xWM63	W			
xWM64	W			
xWM65	W 0065	Refrigeration dryer external shut-off	2.T	Refrigeration dryer: Shut-off via external contact
xWM66	W			
xWM67	W 0067	Condensate drain reservoir	2.2	Tank: Condensate drain faulty
xWM68	W 0068	Condensate drain filter	2.3	Filter: Condensate drain faulty
xWM69	W 0069	Error operation without RD → Call service!	2.T	Error operation without refrigeration dryer is activ. The compressor is running without dryer for a certain time.

1. Compressor ==>>> Master Controller

Table 1.2 : Warning Messages and Maintenance Messages

* Summary of SC2 messages

SC 2 JSON- RPC	Message		Availability	Description / possible cause
	W M ID	Warning Maintenance Text (SC2 Display, except *)		
BDSnd			3 2.x 1	depends on use of the function at SC 2 depends on model type or options, see table 3 always
xWM70	W 0070	Refrigeration dryer T↑	2.T	Refrigeration dryer: Compressed air temperature too high.
xWM71	M			
xWM72	W 0072	RD condensate drain	2.T	Refrigeration dryer: Condensate drain is faulty
xWM73	W 0073	External message 1	3	External binary message 1
xWM74	W 0074	External message 2	3	External binary message 2
xWM75	W 0075	External message 3	3	External binary message 3
xWM76	W 0076	External message 4	3	External binary message 4
xWM77	W 0077	External message 5	3	External binary message 5
xWM78	W 0078	External message 6	3	External binary message 6
xWM79	W			
xWM80	W			

1. Compressor ==>>> Master Controller

Table 1.3 : Operational messages

* Summary of SC2 messages

SC 2 JSON- RPC	Message		Availability	Description / possible cause
BDSnd	ID	Text (SC2 Display, except *)		
xOM1	0001	Delayed start (...s) activ	3	The compressor starts ...s delayed after power on.
xOM2				
xOM3				
xOM4				
xOM5				
xOM6				
xOM7				
xOM8				
xOM9	0009	Compressor on	1	Motor is running or ready for operation
xOM10	0010	Controller on	1	LED at key "I" on
xOM11	0011	Cold start release	3	Cold start is enabled
xOM12				
xOM13				
xOM14				
xOM15				
xOM16				
xOM17	0073	External message 1	3	External binary message 1
xOM18	0074	External message 2	3	External binary message 2
xOM19	0075	External message 3	3	External binary message 3
xOM20	0076	External message 4	3	External binary message 4
xOM21	0077	External message 5	3	External binary message 5
xOM22	0078	External message 6	3	External binary message 6
xOM23				
xOM24				
xOM25				
xOM26				
xOM27	0027	Power OFF → ON	1	It is necessary to switch power supply of SC2 off and on.
xOM28	0028	Dynamic Motor T↑	2.1	Control mode DYNAMIC: The temperature of the compressor motor is too high.
xOM29				
xOM30	0030	Voltage restored	1	SC2 has been feed with supply voltage.

1. Compressor ==>>> Master Controller

Table 1.3 : Operational messages

* Summary of SC2 messages

SC 2 JSON- RPC	Message		Availability	Description / possible cause
	BDSnd	ID Text (SC2 Display, except *)		
xOM31			3	depends on use of the function at SC 2
xOM32			2.x	depends on model type or options, see table 3
			1	always
xOM33	0033	Machine report	3	Life sign has been sent.
xOM34				
xOM35				
xOM36				
xOM37				
xOM38				
xOM39				
xOM40	*	IOSlot Undervoltage error	1	Undervoltage error at SC 2 internal IOM.

1. Compressor ==>>> Master Controller

Table 1.4 : Binary Signals

depends on use of the function at SC 2
depends on model type or options, see table 3
always

3
2.x
1

SC 2 JSON- RPC BDSnd	Description	Availability	Signal	
			"1"	"0"
xBS0	Controller on (LED at key "I" on) + Remote on + Bus (Receive)	1	yes	no
xBS1	Motor running	1	yes	no
xBS2	Load run	1	yes	no
xBS3	Idle	1	yes	no
xBS4	Local mode (LED at key "Remote" off)	1	Remote LED off	Remote LED on
xBS5	Controller on (LED at key "I" on)	1	yes	no
xBS6	Group alarm	1	yes	no
xBS7	Group warning or group maintenance	1	yes	no
xBS8	Compressor ready for load	1	yes	no
xBS9	Compressor ON (Motor is running or ready for operation)	1	yes	no
xBS10				
xBS11				
xBS12				
xBS13				
xBS14	Group maintenance (maintenance messages only)	1	yes	no
xBS15	Group warning (warning messages only)	1	yes	no
xBS16	Mode <i>compressor ON from "I" key only</i> is active	3	yes	no
xBS17	Mode <i>compressor ON from "I" key and clock</i> is active	3	yes	no
xBS18				
xBS19				
xBS20	Mode <i>compressor ON from "I" key and external contact</i> is active	3	yes	no
xBS21	Mode <i>compressor ON from "I" key and clock or external contact</i> is active	3	yes	no
xBS22	Mode <i>compressor ON from "I" key and bus master</i> is active	3	yes	no
xBS23	Mode compressor ON (xBS16 - xBS22) set from	3	Master	vor Ort
xBS24	DUAL control mode is active	3	yes	no

1. Compressor ==>>> Master Controller

Table 1.4 : Binary Signals

depends on use of the function at SC 2
depends on model type or options, see table 3
always

SC 2 JSON- RPC	Description	Availability	Signal	
			"1"	"0"
BDSnd				
xBS25	QUADRO control mode is active	3	yes	no
xBS26	VARIO control mode is active	3	yes	no
xBS27	DYNAMIC control mode is active	2.1	yes	no
xBS28	Continuous control mode is active	3	yes	no
xBS29				
xBS30				
xBS31	Control mode (xBS24 - xBS28) set from	3	bus master	local
xBS32				
xBS33				
xBS34				
xBS35				
xBS36	Mode <i>load-idle signal from external contact</i> is active	3	yes	no
xBS37				
xBS38	Mode <i>load-idle signal from bus master</i> is active	3	yes	no
xBS39	Mode <i>load-idle signal</i> (xBS32 - xBS47) set from	3	bus master	local
xBS40	Mode <i>load-idle signal from pressure regualtor, pA permanent</i> is active	3	yes	no
xBS41	Mode <i>load-idle signal from pressure regualtor, pB permanent</i> is active	3	yes	no
xBS42	Mode <i>load-idle signal from pressure regualtor, pA/pB change via clock</i> is active	3	yes	no
xBS43	Mode <i>load-idle signal from pressure regualtor, pA/pB change via timer</i> is active	3	yes	no
xBS44				
xBS45	Mode <i>load-idle signal from pressure regualtor, pA/pB change via external contact</i> is active	3	yes	no
xBS46	Mode <i>load-idle signal from pressure regualtor, pA/pB change via SC2 master (Ethernet)</i> is active	3	yes	no
xBS47				
xBS48				
xBS49				
xBS50				
xBS51	Automatic restart after power on	1	activated	deactivated
xBS52				
xBS53				

1. Compressor ==>>> Master Controller

Table 1.4 : Binary Signals

depends on use of the function at SC 2
depends on model type or options, see table 3
always

SC 2 JSON- RPC	Description	Availability	Signal	
			"1"	"0"
BDSnd				
xBS54				
xBS55	Message acknowledgement by bus master	1	activated	deactivated
xBS56				
xBS57				
xBS58				
xBS59				
xBS60				
xBS61				
xBS62				
xBS63				

2. Master Controller ==>>>> Compressor

Table 2: Process data

1) write both values at a time
 2) write both values at a time
 * negative value

3 see associated table (2.x)
 2.x see table 3
 1 always

SC 2 JSON- RPC	Description		Usability	Unit	Format
BDRcv					
	Receive bit xBS15 - xBS0	Table 2.1	3	binary	BOOL
	Receive bit xBS31 - xBS16	Table 2.1	3	binary	BOOL
	Receive bit xBS47 - xBS32	Table 2.1	3	binary	BOOL
	Receive bit xBS63 - xBS48	Table 2.1	3	binary	BOOL
iDV0	Setpoint pressure pA switching point 1) <i>if xBS68=1</i>		1	mbar	INT
iDV1	Setpoint pressure pA switching difference * 1) <i>if xBS68=1</i>		1	mbar	INT
iDV2	Setpoint pressure pB switching point 2) <i>if xBS70=1</i>		1	mbar	INT
iDV3	Setpoint pressure pB switching difference * 2) <i>if xBS70=1</i>		1	mbar	INT
iDV4	Speed <i>if xBS48=1</i>		2.E	1/min	INT
iDV5					
iDV6					
iDV7					
iDV8					
iDV9	Actual system pressure at external air receiver pN (pressure regulation) <i>if xBS77=1</i>		1	mbar	INT
iDV10					
iDV11					
iDV12					
	Receive bit xBS79 - xBS64	Table 2.1	3	binary	BOOL
	Receive bit xBS95 - xBS80	Table 2.1	3	binary	BOOL
iMasterMs gHeader	Message header master		1		WORD
iDV13					
iDV14					
iDV15					
iDV16					
iDV17					
iDV18					
iDV19					
iDV20					
iDV21					
iDV22					
iDV23					
iMasterMs gFooter	Message footer master		1		WORD

2. Master Controller ==>>>> Compressor**Table 2.1 : Receive bit assignment**

SC 2 JSON- RPC BDRcv	Description	Usability 2.x 1	see table 3 always		
			Signal		
			"1"	"0"	Kind
xBS0	SAM connection	1	yes	no	duration
xBS1					
xBS2					
xBS3	Continue drive motor running (if xBS0=1)	1	continue running	no bearing	duration
xBS4					
xBS5					
xBS6					
xBS7					
xBS8					
xBS9					
xBS10					
xBS11					
xBS12					
xBS13					
xBS14					
xBS15					
xBS16	Mode compressor ON from "I" key only	1	yes	local	duration
xBS17	Mode compressor ON from "I" key and clock	1	yes	local	duration
xBS18					
xBS19					
xBS20	Mode compressor ON from "I" key and external contact	1	yes	local	duration
xBS21	Mode compressor ON from "I" key and clock or external contact	1	yes	local	duration
xBS22	Mode compressor ON from "I" key and bus master	1	yes (ON/OFF via xBS23)	local	duration
xBS23	Compressor ON / OFF by bus master (if xBS22 = 1)	1	ON	OFF	duration
xBS24	DUAL control mode	1	DUAL	local if xBS24 and xBS25 and xBS26 and xBS27 and xBS28 is "0"	duration
xBS25	QUADRO control mode	1	QUADRO		duration
xBS26	VARIO control mode	1	VARIO		duration
xBS27	DYNAMIC control mode	2.1	DYNAMIC		duration
xBS28	Continuous control mode	1	Continuous		duration
xBS29					
xBS30					
xBS31					
xBS32					

2. Master Controller ==>>>> Compressor**Table 2.1 : Receive bit assignment**

SC 2 JSON- RPC	Description	Usability	Signal		
			"1"	"0"	Kind
BDRcv					
xBS33					
xBS34					
xBS35					
xBS36	Mode load-idle signal from external contact	1	yes	local	duration
xBS37					
xBS38	Mode load-idle signal from bus master	1	yes (load/idle via xBS39)	local	duration
xBS39	Load run by bus master (if xBS38 = 1)	1	load run	idle	duration
xBS40	Mode load-idle signal from pressure regualtor, pA permanent	1	yes	local	duration
xBS41	Mode load-idle signal from pressure regualtor, pB permanent	1	yes	local	duration
xBS42	Mode load-idle signal from pressure regualtor, pA/pB change via clock	1	yes	local	duration
xBS43	Mode load-idle signal from pressure regualtor, pA/pB change via timer	1	yes	local	duration
xBS44					
xBS45	Mode load-idle signal from pressure regualtor, pA/pB change via external contact	1	yes	local	duration
xBS46	Mode load-idle signal from pressure regualtor, pA/pB change via SC2 master (Ethernet)	1	yes	local	duration
xBS47					
xBS48	Set speed	2.E	Master (iDV4)	local	duration
xBS49					
xBS50					
xBS51					
xBS52					
xBS53					
xBS54	Enable message acknowledgement by bus master	1	yes (ack. via xBS55)	local	duration
xBS55	Acknowledge signal for alarms and warnings (if xBS54=1)	1	L→H: acknowledge		flank
xBS56	Enable compressor autostart by bus master	1	yes (set via xBS57)	local	duration
xBS57	Compressor restart (if xBS56=1)	1	yes	no	duration
xBS58					
xBS59					
xBS60					
xBS61					
xBS62					
xBS63					
xBS64					
xBS65					
xBS66					
xBS67					

2. Master Controller ==>>>> Compressor**Table 2.1 : Receive bit assignment**

SC 2 JSON- RPC	Description	Usability	Signal		
			"1"	"0"	Kind
BDRcv					
xBS68	Set pressure pA (switching point and switching difference) via	1	bus master (iDV0+iDV1)	local	duration
xBS69					
xBS70	Set pressure pB (switching point and switching difference) via	1	bus master (iDV2+iDV3)	local	duration
xBS71					
xBS72					
xBS73					
xBS74					
xBS75					
xBS76					
xBS77	Get actual system pressure at external air receiver pN (pressure regulation) via	1	bus master (iDV9)	local	duration
xBS78					
xBS79					
xBS80					
xBS81					
xBS82					
xBS83					
xBS84					
xBS85					
xBS86					
xBS87					
xBS88					
xBS89					
xBS90					
xBS91					
xBS92					
xBS93					
xBS94					
xBS95					

Table 3: Availability of signals for Standard machines

This table covers standard machines. In case of doubt see wiring diagram of the machine.

Avalability	AIRBOX	AIRBOX CENTER	i.Comp-G	i.Comp Tower	Description
1	●	●	●	●	see Table 1.x and 2.x
2.1	2.C	2.C	2.C	2.C	Motor temperature PT100
2.2	x	●	x	●	Condensate drain reservoir
2.3	Option F1/F7	Option F1/F7	Option F1/F7	Option F1/F7	Condensate drain filter
2.A	●	●	x	x	AIRBOX only
2.E	x	x	●	●	i.Comp only (Machine with speed-controlled EC motor)
2.C	C-Maschine, siehe Stromlaufplan der Maschine				
2.T	x	●	x	●	Refrigeration dryer only
3	depends on use of the function at SC 2				

x = not available

● = available

T = machine with dryer